# Thanuish Kumar Sathiaraj

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*M.Sc. Computer Science @ University of Stuttgart | 3D Computer Vision & AR (multi-camera 3D reconstruction @ Max Planck Institute) | Working Student, 20h/week*

## WORK EXPERIENCE

**Student Research Assistant - Computer Vision** Nov 2025 - Apr 2026
*Max Planck Institute for Intelligent Systems (Perceiving Systems, Data Team) - Tübingen, Germany*

- Supported PhD research on audio-driven 3D face reconstruction (generating 3D faces from voice and audio input), collecting and preparing multimodal audio + 3D capture data for training and evaluation.
- Contributed to computer-vision pipelines and a real-time viewer for 3D body reconstruction using multi-camera setups and SMPL-X body models, including keypoint tracking and multi-perspective 3D representation.
- Calibrated and operated a multi-camera motion-capture system (Vicon Nexus) under marker-based protocols, ensuring high-fidelity motion tracking data for 3D body reconstruction research.
- Operated and managed a 4D body scanner for full-body captures, including complex sessions; processed and cleaned raw capture data to reduce noise artifacts and improve tracking consistency.

**Machine Learning Engineer Intern** Apr 2024 - Sep 2024
*Shiash Info-Solutions*

- Developed an end-to-end convolutional neural network (CNN) image-classification pipeline in Python (TensorFlow/Keras), from data collection and cleaning through evaluation - an assigned project delivered independently under senior guidance.
- Applied data augmentation, normalization, and dropout regularization to reduce overfitting; optimized performance with Bayesian hyperparameter tuning for higher generalization accuracy.

## PROJECTS

**VisionARyMenu - AR Restaurant Menu (Android Port)** 2026 - ongoing
*Unity, C#, ARCore, AR Foundation, ARKit (source), glTFast, ZXing.Net, UnityWebRequest (JSON/REST), NUnit*

- Independently ported an iOS/ARKit AR menu application to Android, integrating ARCore and AR Foundation for plane detection and 3D model placement.
- Built a QR-code-triggered remote menu system: decodes table QR codes with ZXing.Net against live camera frames, fetches a restaurant-specific JSON catalog over HTTP (UnityWebRequest), and renders a tappable AR-ready menu - one app build serves many restaurants without hardcoded content.

**Multimodal Deepfake Detection System (Video, Audio, Image)** Aug 2023 - Apr 2024
*Python, OpenCV, TensorFlow, Keras, MTCNN, Inception-ResNet, LSTM, SHAP, Grad-CAM, Librosa, SVM, Random Forest*

- Engineered a cascading detection pipeline (MTCNN face detection to Inception-ResNet feature extraction to KNN classification) for image/video deepfakes, improving detection accuracy 12.4% over standalone models.
- Built an audio deepfake detection module: 20+ hours of speech from ASVspoof, WaveFake and Fake-or-Real datasets, MFCC/spectrogram features via Librosa, dual-stage XAI-CNN + LSTM pipeline with Grad-CAM/SHAP interpretability - 92.1% accuracy with 5-fold cross-validation.

**Helmet Detection & License Plate Recognition**
*YOLOv6, Python, OpenCV, Tesseract/EasyOCR*

- Trained and deployed YOLOv6 for multi-class real-time object detection (motorcycle, rider, helmet status) in a live inference pipeline.
- Built an ROI-cropping and OCR pipeline (Tesseract/EasyOCR) for license-plate extraction; iterated on design parameters for robustness in real-world conditions.

***Further projects:*** *Market Sentinel (multi-agent LangGraph system), Sentiment Transfer (LLM research), Code Grabber (Python OCR)*

## SKILLS

**Computer Vision & 3D:** 3D Reconstruction, Multi-Camera Systems, Camera Calibration, Keypoint Tracking, SMPL-X Body Model, Vicon Nexus, 4D scanning, OpenCV, MTCNN, YOLOv6, OCR
**AR & Mobile 3D:** Unity, C#, ARCore, AR Foundation, glTFast (glTF/GLB), ZXing.Net
**Deep Learning:** TensorFlow, Keras, PyTorch, CNN, LSTM, Inception-ResNet (transfer learning), SHAP, Grad-CAM
**Programming & Tools:** Python, C#, SQL, Bash, Git/GitHub, Docker, Linux, NUnit, Jupyter
**Spoken Languages:** English (C1 - Proficient), German (A2 - currently studying)

## EDUCATION

**M.Sc. Computer Science** Apr 2025 - Jul 2027 (expected)
*University of Stuttgart - Stuttgart, Germany*

**B.E. Computer Science & Engineering** Aug 2020 - Jun 2024
*Anna University*

## INTERESTS

Dancing, Chess, Boxing, Bouldering, Hiking, Gaming, Travelling